Cement composition and hydration

Notes from the lecture slides “Module 1b – Cement composition and Hydration”

In this module

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1 Oxide composition

Chemical analysis reports cement as **oxides**. Cement chemists abbreviate the four main ones:

Oxide	Symbol	Typical content in OPC
CaO (lime)	**C**	60–67%
SiO_2 (silica)	**S**	17–25%
Al_2O_3 (alumina)	**A**	3–8%
Fe_2O_3 (iron oxide)	**F**	0.5–6%
MgO (magnesia)	M	0.5–4%
SO_3 (sulfur trioxide, from gypsum)	$\bar{S}$	2.0–3.5%
Alkalis, expressed as Na_2O	N, K	0.3–1.2%

Other symbols used in hydration chemistry: **H** = H_2O, **CH** = $Ca(OH)_2$, and $\bar{\mathbf{S}}$ = SO_3, so gypsum ($CaSO_4 \cdot 2H_2O$) is written $\mathbf{C\bar{S}H_2}$.

The oxides do not exist separately in cement. In the kiln they combine into **four main compounds**:

Compound	Formula	Short form	Mineral name	Typical content
Tricalcium silicate	$3CaO{\cdot}SiO_2$	**C_3S**	Alite	45–60%
Dicalcium silicate	$2CaO{\cdot}SiO_2$	**C_2S**	Belite	15–30%
Tricalcium aluminate	$3CaO{\cdot}Al_2O_3$	**C_3A**	Aluminate	6–12%
Tetracalcium aluminoferrite	$4CaO{\cdot}Al_2O_3{\cdot}Fe_2O_3$	**C_4AF**	Ferrite	6–8%

Gypsum (about 4%) and additives (up to 5%) are added at grinding.

> **From elements to hydration products**
>
> Elements (Ca, Si, Al, Fe, O) → **oxides** (CaO, SiO_2, Al_2O_3, Fe_2O_3) → **cement compounds** (C_3S, C_2S, C_3A, C_4AF) → **Portland cements** of various types → **hydration products** (C-S-H gel and $Ca(OH)_2$).

2 Bogue equations: from oxides to compounds

The **Bogue equations** convert the oxide analysis (in %) into the approximate (*potential*) compound composition.

Bogue equations (valid when A/F ≥ 0.64)

- $\%\ C_3S = 4.071\ C - 7.600\ S - 6.718\ A - 1.430\ F - 2.852\ SO_3$
- $\%\ C_2S = 2.867\ S - 0.7544\ (\%\ C_3S)$
- $\%\ C_3A = 2.650\ A - 1.692\ F$
- $\%\ C_4AF = 3.043\ F$

Here C, S, A, F and SO_3 are the percentages of CaO, SiO_2, Al_2O_3, Fe_2O_3 and SO_3.

Check the coefficient

The slide shows the silica coefficient in the C_3S equation as 7.700. The standard Bogue equation uses **7.600**; use 7.600 unless your teacher says otherwise.

2.1 Assumptions (and why Bogue values are only approximate)

1. **No secondary or minor phases are present.** This is inaccurate; real clinker has impurities in every phase.
2. **A/F > 0.64.** (For lower ratios a different set of equations is used.)
3. **All aluminoferrite is present as C_4AF.** Not true; ferrite is a series of solid solutions of varying composition.
4. **The system is in equilibrium**, i.e. the clinker cooled slowly enough for all reactions to finish. Real clinker is cooled fast, and some melt freezes as glass.
5. **Free lime is ignored.**

Because of these assumptions, Bogue values are called **potential** compound contents. The actual amounts are found by methods such as X-ray diffraction (see 3.3).

Worked example

An OPC has CaO = 63%, SiO_2 = 20%, Al_2O_3 = 6%, Fe_2O_3 = 3% and SO_3 = 2.5%. Find its compound composition.

A/F = 6/3 = 2.0 ≥ 0.64, so the equations apply.

- $C_3S = 4.071(63) - 7.600(20) - 6.718(6) - 1.430(3) - 2.852(2.5) = 256.47 - 152.00 - 40.31 - 4.29 - 7.13 =$ **52.7%**
- $C_2S = 2.867(20) - 0.7544(52.74) = 57.34 - 39.79 =$ **17.6%**
- $C_3A = 2.650(6) - 1.692(3) = 15.90 - 5.08 =$ **10.8%**
- $C_4AF = 3.043(3) =$ **9.1%**

Total of the four compounds ≈ 90%; the rest is gypsum, MgO, alkalis and minor components.

3 Clinker phases

3.1 What clinker looks like under a microscope

A typical Portland cement clinker contains:

- **large, angular, irregular crystals of alite (C_3S)**,
- **smaller, more rounded crystals of belite (C_2S)**, and
- a **groundmass (interstitial phase)** between them made of the *flux* phases: a mixture of ferrite and C_3A solid solutions that formed from the melt.

3.2 Role of each compound

Compound	Reactivity and heat	Contribution	Notes
C_3S (alite)	High reactivity because of its irregular crystal structure; **high heat of hydration**	**Early strength** (first 7–14 days)	Alite is impure C_3S with about 1% each of MgO, Al_2O_3 and Fe_2O_3, plus small amounts of Na_2O, K_2O, P_2O_5 and SO_3
C_2S (belite)	Less irregular structure, so **less reactive**; low heat	**Later strength** (after about 14–28 days, continuing for months)	4–6% substituted oxides, mainly Al_2O_3, Fe_2O_3 and K_2O
C_3A	**Most reactive**; **highest heat of hydration**	Very little strength; controls early setting	Pure C_3A is cubic; with impurities other forms (e.g. orthorhombic) occur. Distorted bonds make it highly reactive
C_4AF	Moderate	Little strength	Not a fixed compound: it is the *average* composition of a series of solid solutions

C_3A and C_4AF are usually **intermixed** in the groundmass and hard to tell apart.

3.3 How the composition is determined

- Chemical analysis (gives oxides, then Bogue)
- **X-ray diffraction (XRD)** (identifies and quantifies crystalline phases)

- Optical microscopy
- Scanning electron microscopy with energy-dispersive X-ray analysis (SEM-EDX)
- Electron microprobe analysis
- Selective dissolution
- Thermal analysis

4 Hydration and heat of hydration

Hydration is the chemical reaction of cement with water. It is **exothermic**: the heat released is called the **heat of hydration**.

- The **faster the reaction, the faster the heat is released**.
- The heat evolved depends on the heat of hydration of each compound **and** on the clinker morphology (crystal size, defects).

Compound	Heat of hydration (J/g)
C_3S	502
C_2S	260
C_3A	**867** (highest)
C_4AF	419

So a cement rich in C_3S and C_3A gives off more heat, faster. This is why **low-heat cements** have less C_3S and C_3A and more C_2S.

> **Bogue's observation on the rate of heat release**
>
> About **half** the total heat is released **between 1 and 3 days**, about **three quarters in 7 days**, and **83–91% in 6 months**.

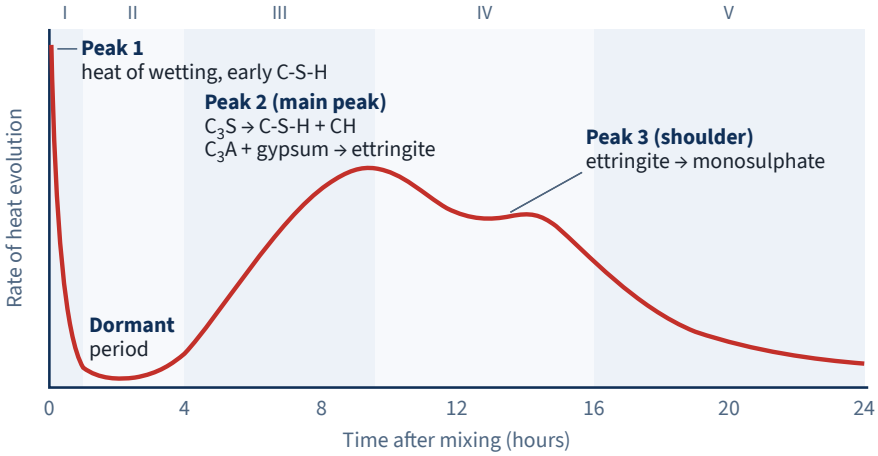
4.1 Measuring heat: calorimeters

Heat of hydration is measured with a **calorimeter**:

- **Adiabatic calorimeter**: no heat is allowed to leave the sample, so the sample's temperature rise is recorded directly.

- **Isothermal calorimeter**: the sample is held at constant temperature and the heat flow out of it (W/kg) is measured.
- **Semi-adiabatic calorimeter** (for example, the one at IIT Madras): a well-insulated sample chamber inside a water tank, with temperature probes and a controller.

4.2 The heat evolution curve



Rate of heat evolution during the first day. Stages: I pre-induction, II dormant (induction), III acceleration, IV deceleration, V slow steady reaction.

Stage	What is happening
Peak 1 (first few minutes)	**Heat of wetting**: rapid dissolution of ions as soon as water touches cement, plus some early C-S-H and ettringite formation
Dormant (induction) period (about 1–3 h)	**Very slow reaction**. The concrete stays plastic and workable, so it can be transported and placed
Peak 2, main peak (about 8–10 h)	**Rapid dissolution and hydration of C_3S → C-S-H + CH**; C_3A + gypsum → ettringite. Initial and final set happen as this peak rises
Peak 3 (shoulder, about 12–20 h)	Gypsum is used up, so **ettringite (AF_t) converts to monosulphate (AF_m)**
After about a day	Slow, diffusion-controlled reaction that continues for months

4.3 Why is there a dormant period?

Several theories exist, but all say that a **layer (of hydration products or of ions) forms on the surface of the cement grains** right after mixing. Further reaction needs water and ions to **diffuse through this layer**, so the reaction slows down.

The dormant period **ends** when:

- the barrier layer **weakens with age** (it breaks or becomes more permeable),
- **diffusion rates increase**, and
- the **ionic strength (concentration) around the grains falls**, as CH and C-S-H start to precipitate.

5 Hydration reactions

5.1 Silicates

C_3S and C_2S react with water to form **calcium silicate hydrate (C-S-H) gel** and **calcium hydroxide (CH)**:

$$2\ C_3S + 6\ H \rightarrow C_3S_2H_3 + 3\ CH$$

$$2\ C_2S + 4\ H \rightarrow C_3S_2H_3 + CH$$

- **C_3S produces three times as much CH as C_2S.** CH adds little strength and is the phase attacked by acids and sulfates, so C_2S-rich cements are more durable.
- C-S-H is the main **strength-giving** product. It has **no fixed composition**: its C/S ratio varies from **1.5 to 2.0**. $C_3S_2H_3$ is only an approximate formula.

5.2 Aluminates without gypsum: flash set

Without gypsum, C_3A reacts almost instantly:

$$2\ C_3A + 21\ H \rightarrow C_4AH_{13} + C_2AH_8$$

- This is the **flash set reaction**: the paste stiffens within minutes with a lot of heat, and cannot be remixed.

- C_2AH_8 is **metastable** and deposits as **hexagonal plates** (like CH). Above about **30 °C** it converts to **cubic hydrogarnet, C_3AH_6**.

5.3 Aluminates with gypsum: ettringite and monosulphate

With gypsum, C_3A forms **ettringite** (AF_t), which coats the C_3A grains and slows their reaction. This is how gypsum controls setting:

$$C_3A + 3\,C\bar{S}H_2 + 26\,H \rightarrow C_6A\bar{S}_3H_{32}\ \text{(ettringite)}$$

- In OPC, **nearly all the sulfate ends up in ettringite**.
- If **C_3A is still left** after the sulfate is used up, it reacts with ettringite to form **monosulphate** (AF_m), $C_4A\bar{S}H_{12}$–18. A commonly written form is $2\,C_3A + C_6A\bar{S}_3H_{32} + 4\,H \rightarrow 3\,C_4A\bar{S}H_{12}$.
- If there is a **large excess of C_3A**, C_4AH_{13} can also form, in solid solution with AF_m.
- **C_4AF** gives similar products, with Fe^{3+} partly replacing Al^{3+} (iron-substituted ettringite with gypsum). The final product depends on how much lime is available.
- The **higher the C_4AF/C_3A ratio, the less ettringite converts to monosulphate**.
- **Ettringite formation is expansive.** Early on, in fresh paste, this is harmless. Ettringite formed later in hardened concrete (for example in sulfate attack) causes cracking.

> **Common short question**
>
> *Why is gypsum added to cement?* To control the very fast reaction of C_3A. Gypsum reacts with C_3A to form ettringite, which coats the C_3A grains and delays their hydration, preventing flash set and giving the dormant period needed for placing concrete.

5.4 Points to ponder

- Heat evolution can only be related correctly to temperature rise if the system is **perfectly insulated**.

- **Water/cement ratio matters**: water has a much higher specific heat than cement, so the more water in the mix, the more heat is needed to raise its temperature.
- Cement contains **highly soluble alkali oxides** (Na_2O and K_2O). They dissolve quickly and make the pore solution **highly alkaline (pH 12–13)**. Cement therefore hydrates in this **pore solution, not in pure water**.

5.5 Effect of w/c on hydration (test data, Prasath 2012)

w/c	Peak heat rate (W/kg)	Time to peak (h)	Dormant period (h)	Total heat at 3 days (kJ/kg)	Peak temp. (°C)	Strength 3 / 7 / 28 days (MPa)
0.40	7.72	5.75	1.5	289	70.2	44.4 / 50.4 / 64.4
0.45	6.19	8.25	3	281	65.9	35.1 / 45.3 / 61.2
0.50	4.77	9.25	3	262	60.3	29.3 / 36.6 / 47.1
0.55	4.40	9	3	242	60.5	22.1 / 27.9 / 36.9
0.60	4.19	9.5	4	227	59.8	15.0 / 24.0 / 29.1

What the table shows: as w/c increases, the peak heat rate, total heat (per kg of mix) and peak temperature fall, the dormant period and time to peak get longer, and **strength drops sharply** at every age.

6 Microstructure of hydrated cement paste

6.1 Inner and outer C-S-H

Inner C-S-H (phenograin C-S-H)	Outer C-S-H (groundmass C-S-H)
Forms **inside the original boundary** of the cement grain	Forms in the **water-filled space** between grains
Dense (high-density C-S-H)	**Less dense, more porous** (low-density C-S-H)
Contains **less Al and S**	Contains **more Al and S** (mixed with AF_m)
Forms around **large** grains	Forms from **small** grains

Low-density (outer) C-S-H forms mainly during the main heat peak; high-density (inner) C-S-H forms later, during the slow diffusion stage.

6.2 What the micrographs show

- **C-S-H** looks gel-like or fibrous/spiky ("hedgehog" form on C_3S grains); it has no well-defined crystal shape.
- **CH** forms large **hexagonal plates** and often appears as a **white rim around aggregates** in back-scattered images.
- **Ettringite** forms **long needles**; **monosulphate** forms **thin hexagonal platelets**.
- In back-scattered SEM images, **bright particles are unhydrated cement**, **grey regions are C-S-H**, and **black regions are pores**.

6.3 Pores in hydrated paste

Hydrated cement paste = **hydration products** (C-S-H, CH, AF_t, AF_m…) + **capillary pores**.

- **Gel pores** are the tiny pores *within* the hydration products (mainly inside C-S-H).
- **Capillary pores** are the remains of the original water-filled space not filled by hydration products.

- According to **Powers**, about **one third of the pore space is gel pores** and the rest is capillary pores.

Pore type	Size	Water held	Affects
Large capillary pores	10 µm – 50 nm	Bulk (evaporable) water	Permeability, strength
Medium capillary pores	50 – 10 nm	Evaporable, moderate menisci	Permeability, strength, shrinkage at high RH
Small gel pores	10 – 2.5 nm	Evaporable, strong menisci	Shrinkage (down to 50% RH)
Gel micropores	2.5 – 0.5 nm	Non-evaporable, no menisci	Shrinkage, creep (35–11% RH)
Interlayer spaces	< 0.5 nm	Non-evaporable, ionic/covalent bonds	Shrinkage, creep (below 11% RH)
Interfacial transition zone (ITZ)	20 – 50 µm	Bulk water	Permeability, strength
Microcracks	50 – over 200 µm	Bulk water	Permeability, strength

For scale: **entrained air bubbles** are about 10 µm – 1 mm, **entrapped air voids** are larger (millimetres), while **interlayer spacing in C-S-H** is about 1 nm.

6.4 Water in cement paste

1. **Capillary water**, in voids larger than about 50 Å (5 nm):
   - *free water* in large voids: removing it causes **no shrinkage**;
   - water held by **capillary tension** in small pores: removing it **causes shrinkage**.
2. **Adsorbed water**: held on the surfaces of hydration products, mainly C-S-H, in several molecular layers. Much of it is lost when RH drops to about 30%, causing **a lot of shrinkage**.
3. **Interlayer water**: held between the layers of C-S-H. Losing it causes **large shrinkage** as the C-S-H structure collapses.

4. **Bound (chemically combined) water**: part of the hydration products; removed only on **ignition**. Also called **non-evaporable water**.

Adsorbed + interlayer water together are called *gel water*.

7 Factors affecting the rate of hydration

1. **Phase composition** of the cement (more C_3S and C_3A → faster).
2. **Amount and form of gypsum**: whether it is dihydrate, hemihydrate or anhydrite (they dissolve at different rates).
3. **Fineness**: finer cement has more surface area, so it reacts faster.
4. **Water/cement ratio**: at high w/c hydration can continue until all cement is used; at low w/c it may stop for lack of water and space.
5. **Curing conditions**: relative humidity strongly affects how far hydration proceeds.
6. **Temperature**: higher temperature speeds up the reaction, although the hydrated structure formed may be different (coarser, less dense).
7. **Chemical admixtures**: e.g. accelerators, retarders (set controllers) and plasticizers.

Quick revision

- Oxides C, S, A, F → compounds C_3S (45–60%), C_2S (15–30%), C_3A (6–12%), C_4AF (6–8%).
- Bogue: $C_3S = 4.071C - 7.600S - 6.718A - 1.430F - 2.852SO_3$; $C_2S = 2.867S - 0.7544C_3S$; $C_3A = 2.650A - 1.692F$; $C_4AF = 3.043F$. Valid for $A/F \geq 0.64$; gives *potential* composition only.
- C_3S: early strength, high heat. C_2S: later strength, low heat. C_3A: highest heat (867 J/g), flash set, little strength. C_4AF: little strength.
- Heat curve: peak 1 (wetting) → dormant period → main peak (C_3S hydration, setting) → shoulder (AF_t → AF_m).
- $2C_3S + 6H \rightarrow C_3S_2H_3 + 3CH$; $2C_2S + 4H \rightarrow C_3S_2H_3 + CH$; $C_3A + 3C\bar{S}H_2 + 26H \rightarrow$ ettringite.

- Gypsum prevents flash set by forming ettringite on C_3A grains. Ettringite is expansive.
- Pore solution pH 12–13 (alkalis). Gel pores ≈ 1/3 of pore space (Powers).
- Water: capillary, adsorbed, interlayer (adsorbed + interlayer = gel water), bound (non-evaporable).

Questions to practise

1. Name the four main compounds of Portland cement, with their formulas and typical percentages. (4 marks)
2. Write the Bogue equations and state their assumptions. Why are Bogue values called "potential" compositions? (6 marks)
3. A cement has CaO 64%, SiO_2 21%, Al_2O_3 5.5%, Fe_2O_3 3.5% and SO_3 2.4%. Estimate its compound composition. (6 marks)
4. Compare the roles of C_3S, C_2S, C_3A and C_4AF in strength development and heat of hydration. (6 marks)
5. Sketch the rate of heat evolution curve for OPC and explain each stage. Why is there a dormant period? (10 marks)
6. Write the hydration reactions of C_3S, C_2S and C_3A (with and without gypsum). (6 marks)
7. What is flash set? How does gypsum prevent it? (3 marks)
8. Distinguish between inner and outer C-S-H. (3 marks)
9. Classify the pores in hydrated cement paste by size and explain how each affects properties. (6 marks)
10. Describe the different types of water held in hydrated cement paste. Which ones cause shrinkage when lost? (5 marks)
11. List the factors that affect the rate of hydration of cement. (5 marks)